Industrial Electronics N5
Power Supplies
Module 2

Learning Outcomes

At the end of this module, learners will be able to:

- explain sine waves using graphic representation, which must indicate frequency, periodic time, peak value, peak-to-peak value, average value, RMS value, and form- and crest factor (FF and CF);
- explain the following using circuit diagrams:- input-and output waveforms the application of a diode in half wave rectification, full wave rectification (two diodes) and full wave rectification (bridge circuit - four diodes).
- explain and calculate of the following half-wave and full-wave diode rectifier concepts:
    - transformer ratio
    - average DC voltage, ripple voltage, ripple factor, peak inverse voltage (PTV) ratings;
- explain, using sketches, what is meant by filter circuits in rectifiers.
- illustrate, using diagrams, the application of half-wave rectifiers with capacitor- and filters, including the following arrangements:
    - two diodes, bridge circuit with four diodes;
- calculate the following in relation to capacitor filters and RC filters respectively:
    - the average- and RMS values, output DC voltage, output RMS ripple voltage, peak inverse voltage;
- draw a circuit diagram that will illustrate the principle of no-load voltage;
- draw a circuit diagram, graphical representation and perform calculations that will illustrate the principle of voltage regulation;
- explain, using suitable circuit diagrams, the application of a diode in full wave voltage doubling circuits, and other voltage multiplication circuits;
- understand, explain, and perform calculations in relation to a Zener diode used as a voltage reference source;
- xplain series- and shunt regulators using suitable circuit- and block diagrams;
- understand and explain the following, using applicable circuit diagrams:
    - basic series voltage regulator control
    - series voltage regulator
    - variable series voltage regulator
    - series regulator with an adjustable reference source
    - series regulator with an amplifier
    - series regulator with overcurrent protection
    - overvoltage protection and a shunt voltage regulator;
- describe and explain the application of the following using suitable diagrams:
    - a three-terminal voltage regulator.
    - a double voltage power supply; and

- describe and explain the application of the following adjustable power supply regulators using block circuit diagrams:
    - a higher than fixed voltage of regulator
    - a lower than fixed voltage of regulator
    - a highly stabilized and adjustable regulator.

2. POWER SUPPLIES

2.1. INTRODUCTION:

A power supply can be defined as a hardware element that supplies power to an electrical device. It obtains power from an electrical outlet and converts the current from AC (alternating current) to DC (direct current), which is what Electrical devices need to operate.

2.2.OVERVIEW

A summary of the equations you covered in **N4** is given below:

A.) SINE WAVE

$$Form\ factor = \frac{V_{RMS}}{V_{ave}} = 1{,}11$$

$$V_{rms} = \frac{1}{\sqrt{2}}(V_m) = 0{,}707V_m$$

$$V_{ave} = \frac{2}{\pi}(V_m) = 0{,}637V_m$$

$$Peak\ factor = \frac{V_m}{V_{rms}}\ 1{,}414$$

$$\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$$

B.) HALF-WAVE RECTIFICATION WITHOUT SMOOTHING

$$V_{dc} = \frac{1}{\pi}(V_m) = 0.318\,V_m$$

$$PIV = V_m$$

$$V_{r(rms)} = 0{,}385V$$

$$Ripple\ factor\ r = \frac{V_{r(rms)}}{V_{dc}} = \frac{0{,}385\,V_m}{0{,}318V_m}\ 1{,}21$$

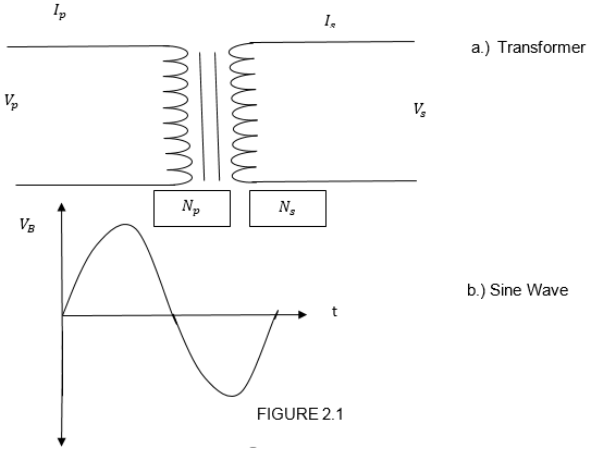
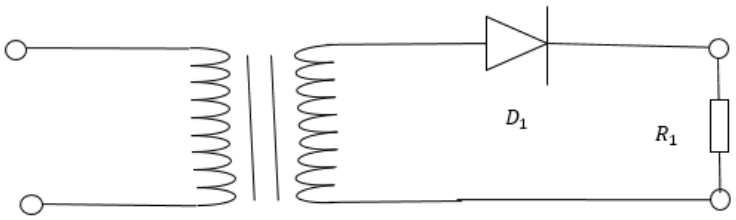


FIGURE 2.1



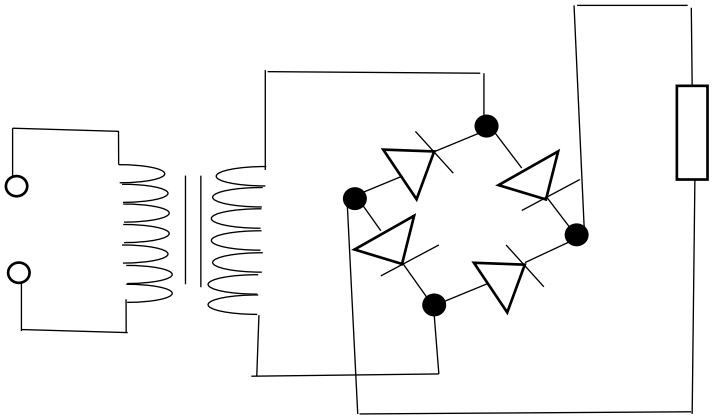
a.) Half -wave rectifier



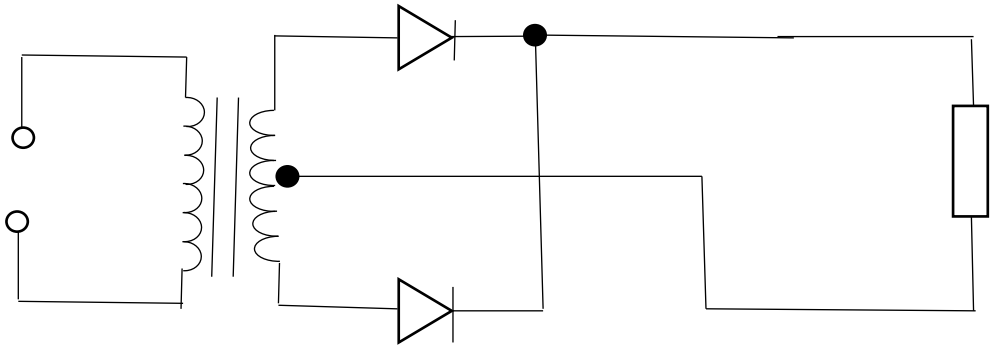
b.) Half Wave Output

FIGURE 2.2

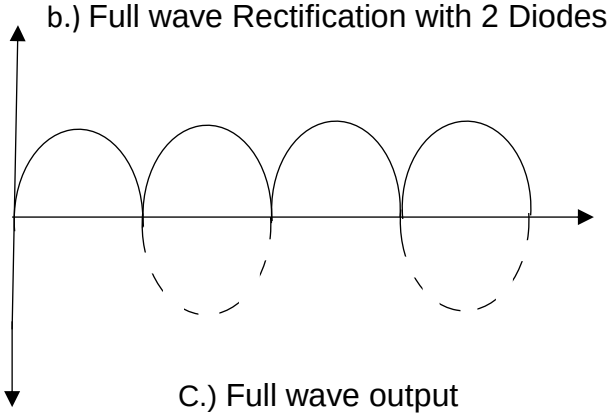
C.) FULL-WAVE RECTIFICATION WITHOUT SMOOTHING



a.) Full wave rectification with 4 Diodes



b.) Full wave Rectification with 2 Diodes



C.) Full wave output

$$V_{DC} = \frac{2}{\pi}(V_m) = 0{,}637V_m$$

PIV $= V_m (For\ 4\ diodes)$

PIV $= 2V_m\ (For\ 2\ diodes)$

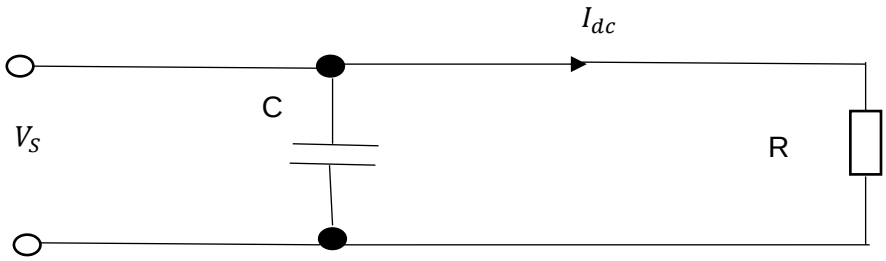
$$V_{r(rms)} = 0{,}305V_m$$

$$r = \frac{V_{r(rms)}}{V_{dc}}$$

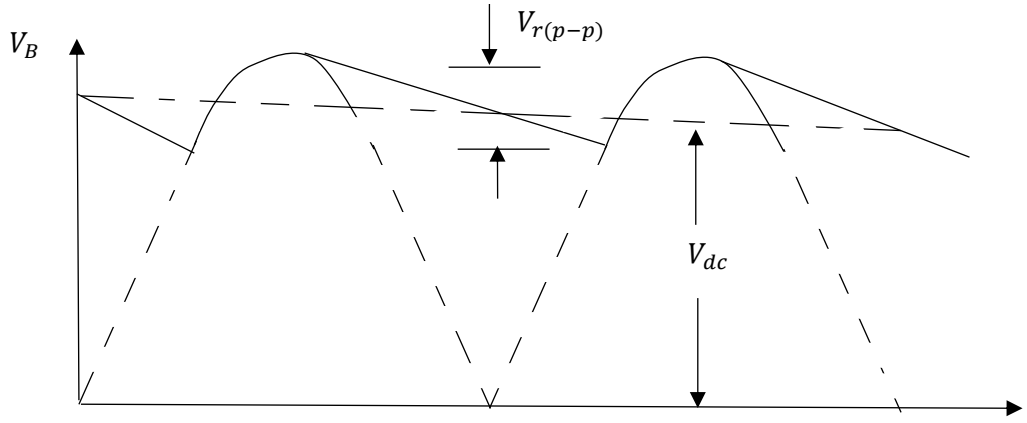
$$= \frac{0{,}305V_m}{0{,}637V_m} =$$

FIGURE 2.3

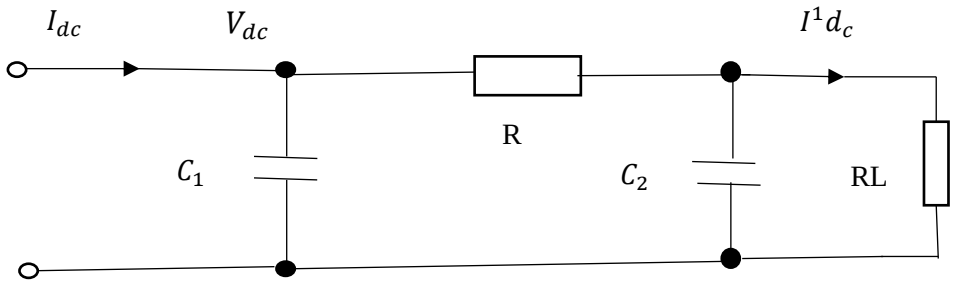
D.) FILTER CIRCUITS



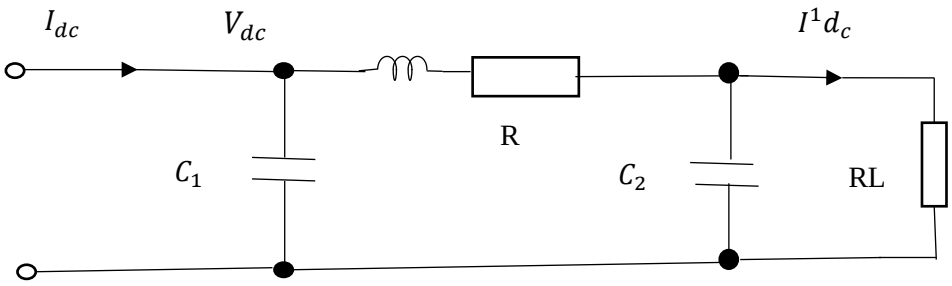
a.) Simple Capacitor Filter



b.) Smoothed Output Waveform



c.) RC π Filter



d.) LC π Filter

$$V_{dc} = V_m - \frac{V_{r(p-p)}}{2}$$

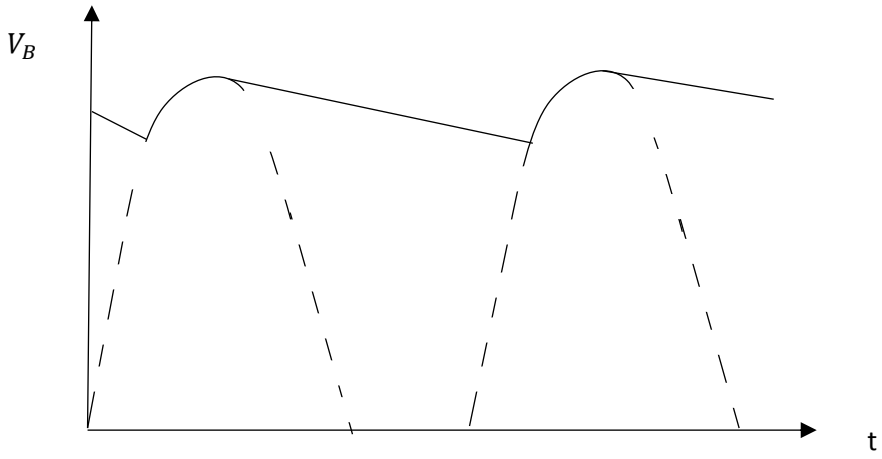
$$V_{r(rms)} = \frac{V_{r(p-p)}}{2\sqrt{3}}$$

$$r = \frac{V_{r(rms)}}{V_{dc}}$$

FIGURE 2.4

E) SIMPLE CAPACITOR CIRCUIT

1.)Half - Wave

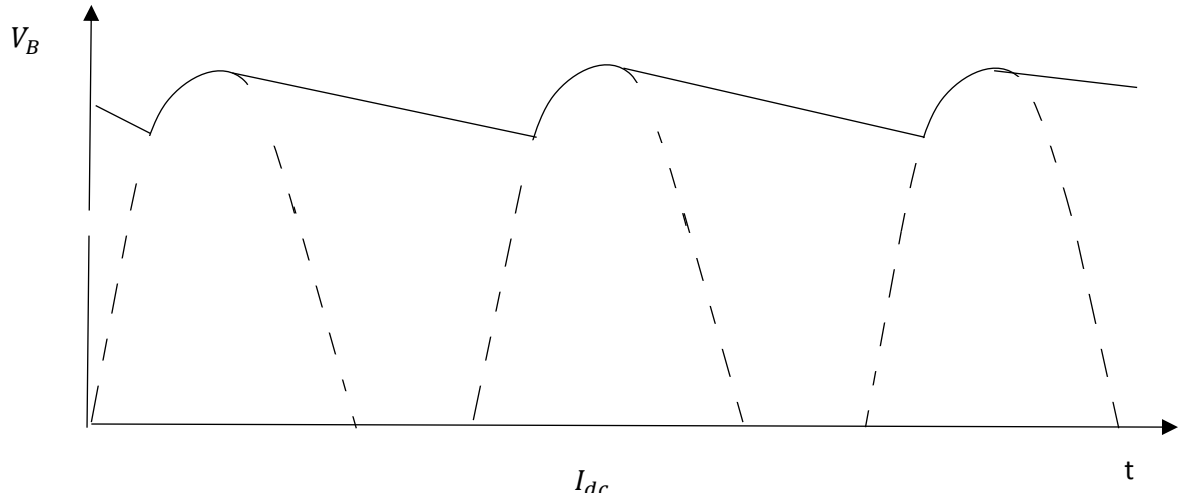


$$V_{dc} = V_m - \frac{I_{dc}}{2fC}$$

$$V_{r(rms)} = \frac{I_{dc}}{2\sqrt{3fC}} = \frac{V_{dc}}{2\sqrt{3fC}\,(RL)}$$

$$r = \frac{I_{dc}}{2\sqrt{3fC}V_{dc}} = \frac{1}{2\sqrt{3fC}(RL)}$$

FIGURE 2.5 Half wave output



$$V_{dc} = V_m - \frac{I_{dc}}{4fC}$$

2.)Full - Wave

$$V_{r(rms)} = \frac{I_{dc}}{4\sqrt{3fC}} = \frac{V_{dc}}{4\sqrt{3fC}\,(RL)}$$

$$r = \frac{I_{dc}}{4\sqrt{3fC}V_{dc}} = \frac{1}{4\sqrt{3fC}(RL)}$$

FIGURE 2.6 Full wave output

F.) π – FILTERS

i.) RC π Filters

$$V^1{}_{dc} = \frac{R_L}{R+R_L}(V_{dc})$$

$$V^1{}_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X^2{}_C}}\,(V_{r(rms)}$$

$$V_{r(rms)where}\ X_C = \frac{1}{2\pi fC}\ for\ half - wave$$

$$X_C = \frac{1}{4\pi fC}\ for\ full - wave$$

$$or\ V^1{}_{r(rms)} = \frac{X_C}{R}\ (V_{r(rms)}\ if\ R > 5\ X_C$$

$$r^1 = \frac{V^1{}_{r(rms)}}{V^1{}_{dc}}\ \ (accurate\ value)$$

$$or\ \ r^1 = (r)(X_c)(\frac{R + R_L}{(R)(R_L)}$$

ii.) LC π Filters

$$V^1{}_{dc} = V_{dc} - (I_{dc})(R_1)$$

$$V^1{}_{r(rms)} = \frac{R_L}{R_L + R_1}(V_{dc}) \quad - Voltage\ divider\ rule$$

$$V^1{}_{r(rms)} = \frac{V_{r(rms)}}{4\pi^2 f^2 LC} \quad for\ half - wave$$

$$V^1{}_{r(rms)} = \frac{V_{r(rms)}}{16\pi^2 f^2 LC} \quad for\ full\ wave$$

2.3 VOLTAGE REGULATION

Voltage regulation is the amount of change the output DC-voltage experiences during varying load conditions. The equation for the percentage of voltage regulation is:

$$\%\,VR = \frac{V_{NL} - V_{FL}}{V_{FL}}\,(100)$$

Watch the video below for more information on voltage: https://youtu.be/IjJWWGPjc-w

Example 2.1

Calculate the percentage voltage regulation of a power source if the ripple factor and ripple voltage without a load is 2% and 0,244V, respectively & the full load voltage is 11,8V.

$$r = \frac{V_{r(rms)}}{V_{dc}} = \frac{0{,}244}{0{,}02} = 12{,}2\ V$$

2.4 VOLTAGE DOUBLER AND TRIPLER

Voltage multipliers are used as soon as a higher voltage is necessary from a transformer secondary voltage. Capacitor filters are used for this. ***However, it cannot maintain a large load current.***

FIGURE 2.7 depicts a half-wave voltage doubler.

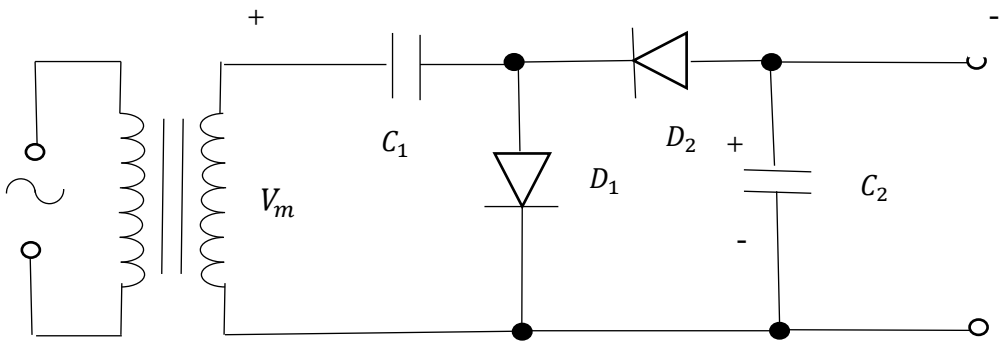


FIGURE 2.7 Half wave voltage doubler

During the positive half-cycle across the transformer, diode D1 conducts while diode D2 is reverse biased. Capacitor C1 charges to a maximum voltage of V_m. Because diode D1 effectively is a short circuit, the voltage drop across capacitor C2 will be zero ($V_{c2} = 0$). During the negative half-cycle across the transformer, diode D2 will conduct while diode D1 is reverse biased. Capacitor C1 is prevented from discharging because diode D1 does not conduct and keeps its charge. Therefore, capacitor C2 charges to a voltage value of

$2V_m$: $V_{C2} = V_m + V_{c1} = 2V_m$.

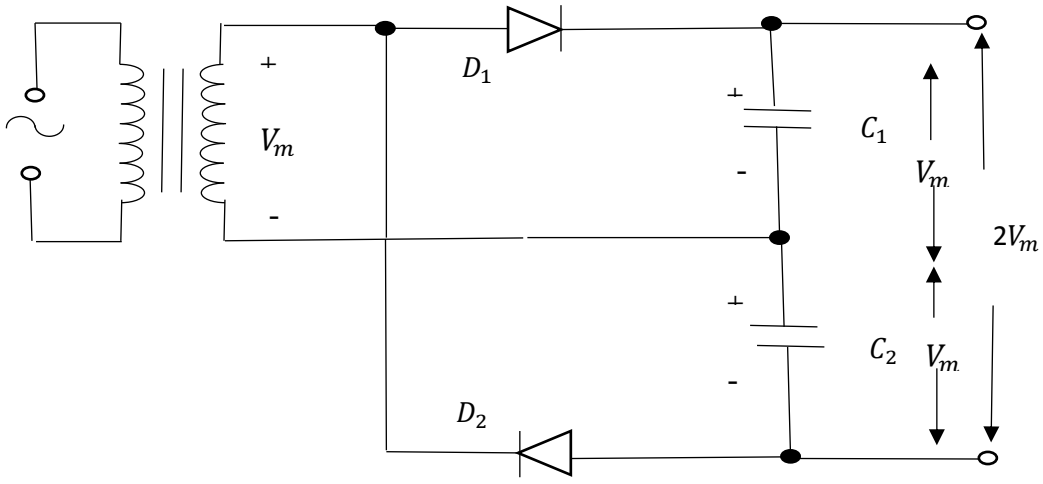


FIGURE 2.8 full- wave voltage doubler

During the positive half-cycle across the transformer, diode D1 conducts, and D2 is off. Capacitor C1 charges to a voltage of V1. During the negative half-cycle, diode D2 will conduct, and diode D1 will be off. Capacitor C2 will then charge to a voltage of V_m. The total voltage drop across both capacitors will therefore be $2V_m$.

A trippler is obtained if the half-wave doubler is expanded with another diode capacitor stage.

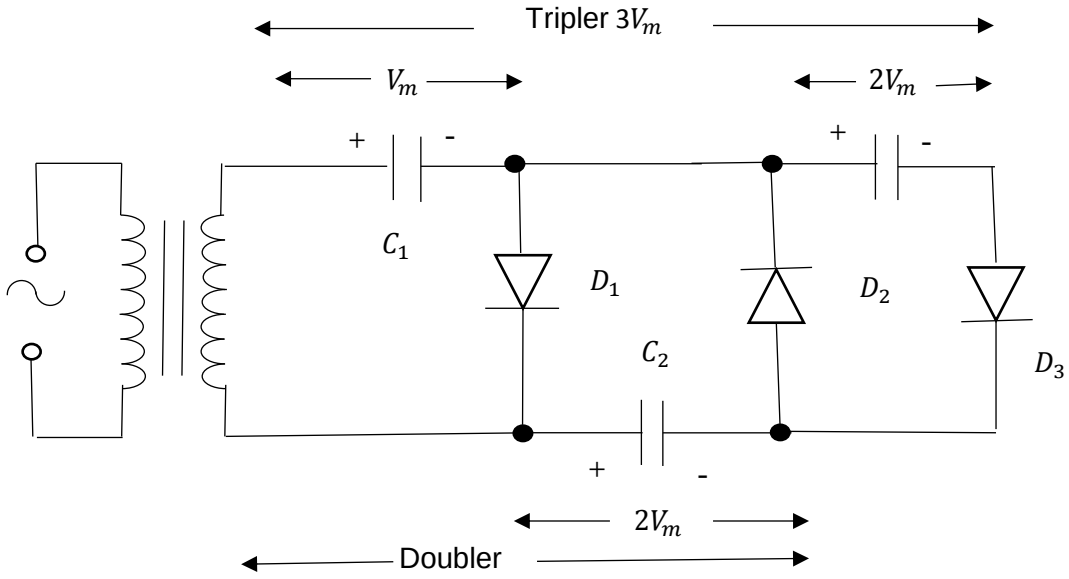


FIGURE 2.9 Voltage Tripler

Capacitor C1 charges to $V_m$through diode D1 during the positive half-cycle. During the negative half-cycle, capacitor C2 charges to $2V_m$. Capacitor C3 charges to a voltage of 2 V11 during the positive half-cycle. The total voltage drops across capacitors C1 and C2; therefore, it is:
$V_{C1} + V_2 = V_m + 2V_m = 3V_m$

The secondary voltage can be enlarged even further by adding similar stages. Every following capacitor is charged to a voltage of $2V_m$A voltage is added to the previous output.

2.5 SERIAL AND SHUNT REGULATORS

A voltage regulator normally consists of five major Elements:

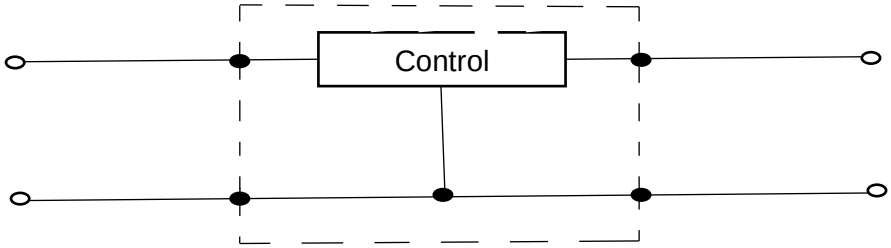
a.) A reference element which provides a stable reference level.

b.) ***A feedback network*** supplies a feedback signal.

c.) ***A comparator*** compares the feedback signal with the reference level and supplies an error signal.

d.) ***An optional protective element*** which protects the control element.

e.) ***A control element*** reacts upon the error signal to keep the output voltage constant during varying load conditions or supply voltage. This element can be a series or shunt regulator.



a.) Serial regulators

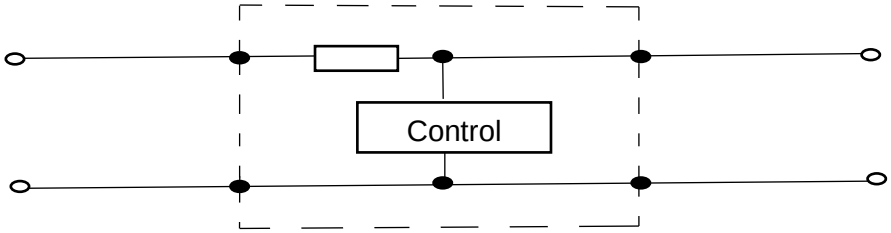


FIGURE 2.10 b.)Serial regulators

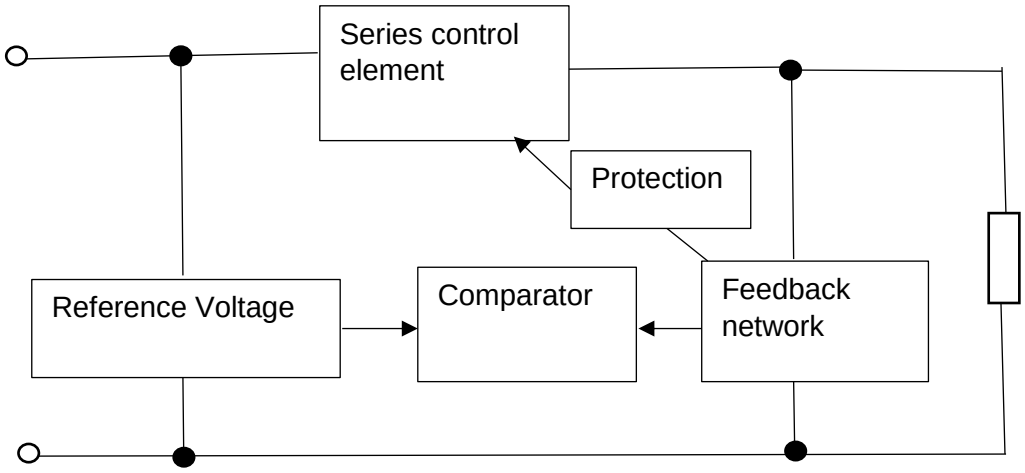


FIGURE 2.11 Block diagram of a series voltage regulator

The function of the control element is ***to regulate the output voltage***. A part of the output voltage is fed back by a voltage distributor and compared with the reference voltage. The difference between these two voltages is then supplied to the control element, which controls the output voltage correspondingly.

The degree of stabilizing during a voltage regulator is called the stabilizing factor (s) and is determined by the following equation:

$$S = \frac{\Delta V_{OUT}}{\Delta V_{in}}$$

2.6 THE VOLTAGE REFERENCE SOURCE

The Zener-diode, obtainable in various voltage and power values, is usually used as a voltage reference source. A Zener-diode will regulate if the input voltage V_{in} is greater than the Zener voltage V_z. A series resistor is therefore required to absorb the voltage difference, which means that:
$V_R = V_{in} - V_z$

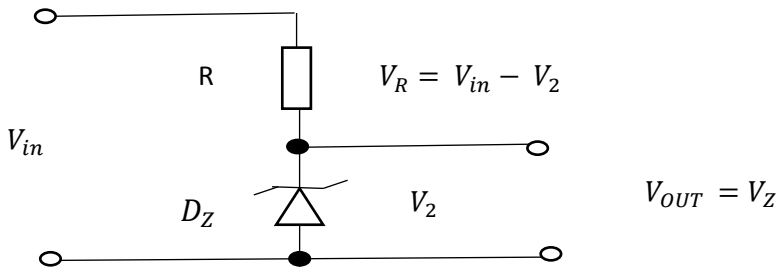


FIGURE 2.12 A Zener-diode reference source

If V_{in} changes, V_R will also change while V_Z remains constant. The Zener-current I_Z will, however, also change. The maximum power rating of the Zener-diode P_Z determines the maximum Zener-current. Therefore, to protect the Zener-diode against excessive current, a minimum value for the series resistor must be calculated using the following equation:

$$R_S = \frac{V_{in(max)} - V_Z}{I_{Z(max)}} \quad , where\ I_{Z(max)} \leq \frac{P_Z}{V_Z}$$

$$and\ P_z = power\ rating\ of\ the\ zener$$

If a load resistor R_L is connected across a Zener diode, the load resistor will cause the current through the Zener diode to drop. As soon as the Zener-current approaches 0, the Zener-diode will stop regulating. This situation can be avoided by calculating a minimum value for the load resistor $R_{L(\min)}$.

$$Thus\ from\ \frac{V_S}{R_{L(\min)}} = \frac{V_{in(\max)} - V_Z}{R_S} \qquad (I_L = I_S)$$

$$R_{L(\min)} = \frac{V_Z}{V_{in(\max)}}\ (R_S)$$

Example 2.2

A voltage reference circuit uses a 10 V, 500 mW Zener diode and is supplied with a 15 V ± I V-dc supply. Calculate the minimum values for both the series and load resistors.

$$I_Z \leq \frac{P_Z}{V_Z} = \frac{500\ \text{mW}}{10\text{V}} = 50\ \text{mA}$$

$$R_{S(min)=} \frac{V_{in(max)} - V_Z}{I_Z} \qquad (R_{L=} \infty)$$

$$= \frac{16 - 10}{0{,}05} = 120\Omega$$

$$R_{L(min)=} \frac{V_Z}{V_{in(max)} - V_Z} (R_S) = \frac{10}{16 - 10}(120) = 200\Omega$$

2.7 SERIES VOLTAGE REGULATORS

The basic circuit of a series regulator in an emitter follower is indicated in **FIGURE 2.13.** The reference voltage V_r of the reference source is applied on the base of the transistor.

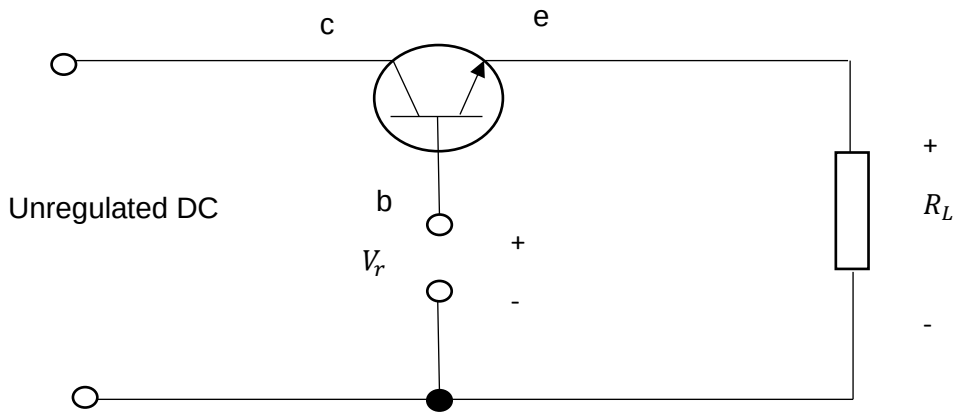


FIGURE 2.13 Basic series control

The purpose of the transistor, which only has a minor voltage drop across the collector-emitter junction, is to increase the current supply rating to the load. It is done by bypassing the voltage reference source. While the transistor is switched on, the output voltage will be $V_{OUT} = V_r - V_{be}$. The output voltage of this circuit is fixed.

If you want to adjust the output voltage, a potentiometer can be connected as indicated **in FIGURE 2.14**. ***The potentiometer P serves as a voltage divider which makes the reference voltage V_r adjustable.***

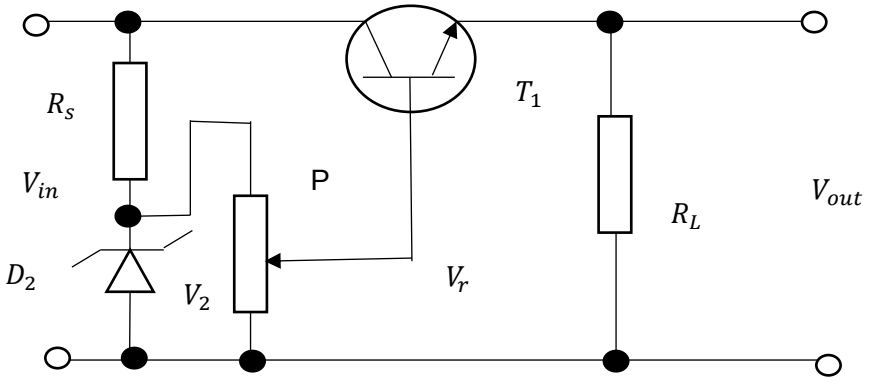


FIGURE 2.14 Series voltage regulator with adjustable reference source

The stabilizing factor of the circuit can be improved by connecting a feedback network with a comparator or a differential amplifier as indicated **in FIGURE 2.15.**

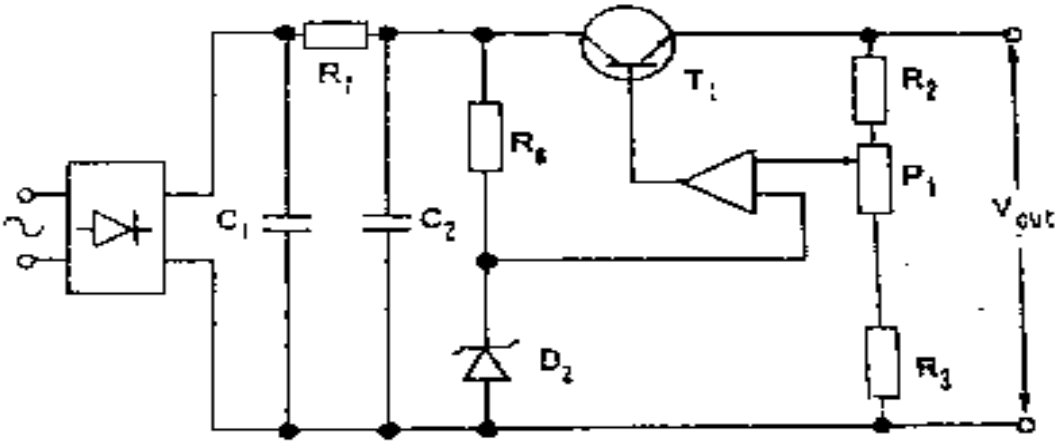


FIGURE 2.15 Series voltage regulator with amplifier

Potentiometer P1, together with R2 and R3, form an adjustable voltage divider network which supplies a feedback voltage V_f from the output voltage V_{out} to the differential comparator. This feedback voltage is continuously compared with the reference voltage V_Z. The difference between the two voltages is amplified as error voltage, where the transistor regulates the output voltage accordingly.

Any increase in output voltage because of an increase in input voltage will produce a higher feedback voltage. The error voltage $(V_Z - V_f)$ increases, and because the output of the amplifier is out of phase with the error voltage, the transistor T_1 is forced to conduct less current. This action causes the output voltage to fall and therefore be stabilized.[1]

2.8 OVER CURRENT PROTECTION

An over current protection circuit can be added to protect both the control transistor T_1 and the load against the excess current. **FIGURE 2.16** indicates this circuit.

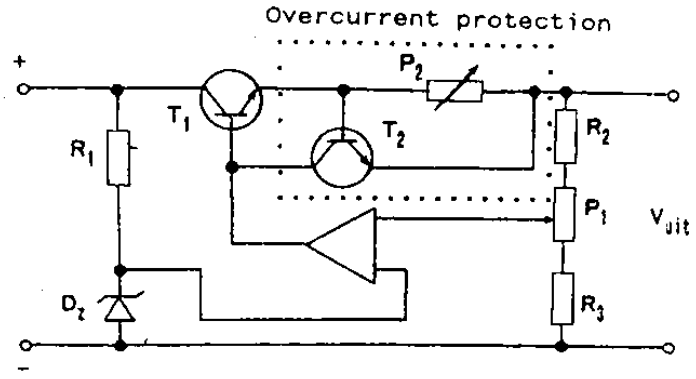


Fig 2.16 Over current protection in a series regulator

Potentiometer P2 is adjusted during a normal load current so that transistor T2 is not biassed forward sufficiently and therefore is off. An increase in load current will cause an increase in voltage drop across P2.

During overload conditions, the increased voltage drop across P2 will switch transistor T2 on to conduct a part of the base current of transistor T1 and limits the increase in load current to a safe value.

The value at which P2 is set determines the maximum load current which may flow. A requirement of P2 is a low resistance with a high-power rating.

2.9 OVER VOLTAGE PROTECTION

Over voltage protection is applied in stable Power supplies that must prevent certain circuits sensitive to excessive supply voltage from damage, such as integrated circuits and colour televisions. FIGURE 2.17 shows an over-voltage protection circuit connected parallel with the load.

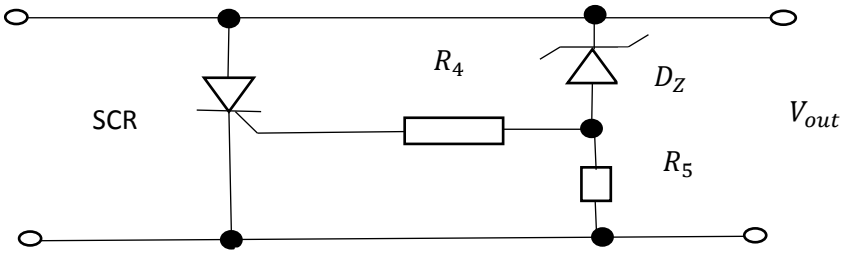


FIGURE 2.17 Over voltage protection

The values of D_Z and R_5 determine the maximum allowable voltage that may appear across the load. Under normal output voltages, the voltage drops across R_5 will be too low to trigger the gate of the SCR, and therefore, the SCR will be off as soon as the maximum allowed voltage is exceeded. The voltage drop across the Zener will stay constant while that across the R_5 increases. This increase is of such an extent that the gate of the SCR is triggered, and the SCR is switched on. Because the SCR causes a short circuit, the output voltage drops to zero. The overcurrent protection circuit can limit the sudden increase in current.

2.10SHUNT VOLTAGE REGULATOR

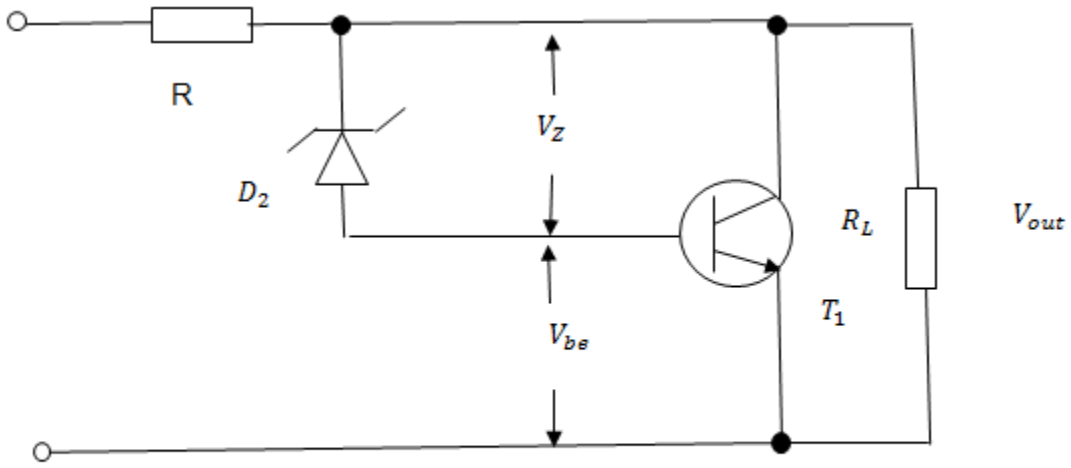


FIGURE 2.18 Shunt voltage regulator

The value of the Zener-diode D_z is nominated in such a way that the bias V_{be} of the transistor T_1 is low enough to keep the transistor in an unsaturated condition. If the supply voltage increases,

the Zener voltage V_Z will remain constant, but the base bias V_{be} will increase. This causes the transistor to be driven to greater saturation to let the collector voltage drop. This fall compensates for the increase in supply voltage, and in this way, the output voltage is stabilised.

2.11 DOUBLE VOLTAGE POWER SUPPLIES

Differential and operational amplifiers commonly require a positive and negative power supply, typically nl +/- 15V. To satisfy this requirement, a double voltage power supply is used.

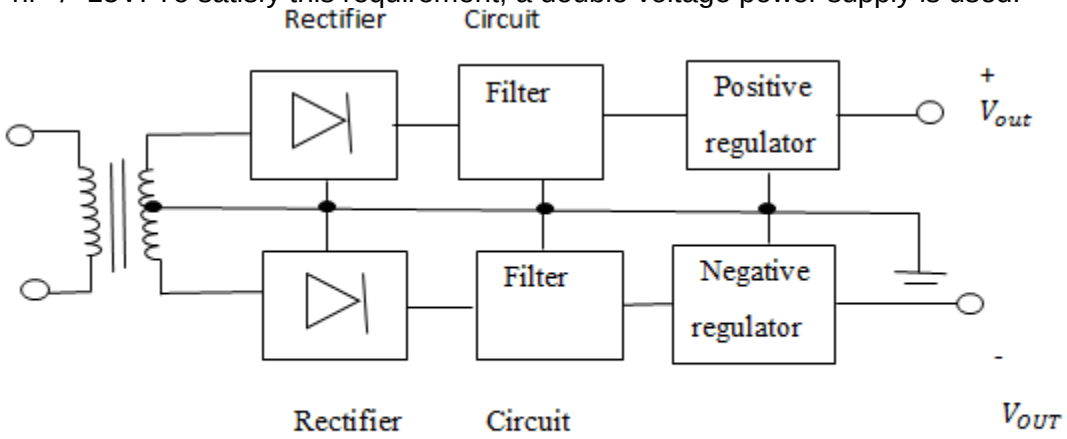


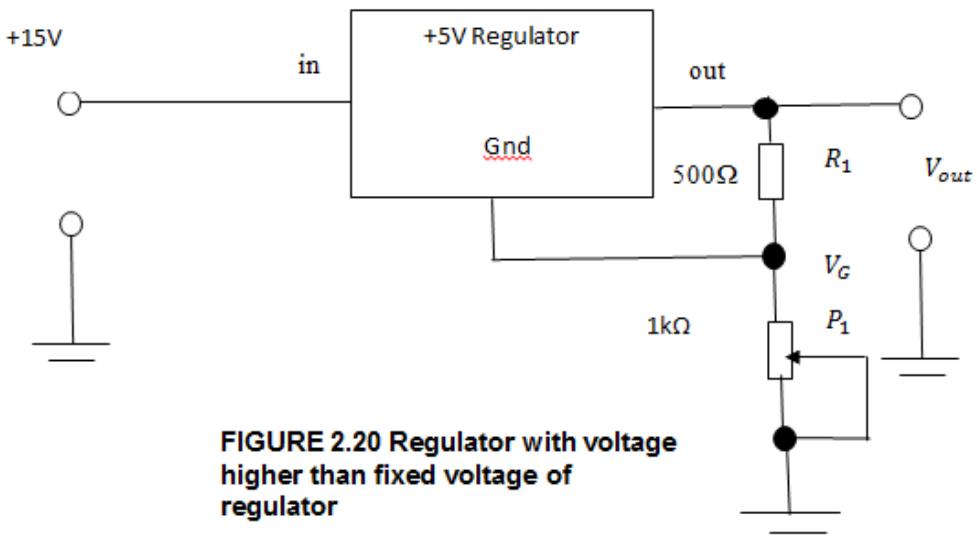
FIGURE 2.19 Double power supply

FIGURE 2.19 indicates that every output voltage, + V_{out} and -V_{out} has its rectifier, filter, and regulator. The common line is connected to the center tap of the secondary winding of the transformer.

2.1.2 ADJUSTABLE POWER SUPPLIES

Fixed voltage power supplies are used for specific circuits. They form part of electronic equipment (radios, televisions, measuring equipment), while adjustable power supplies are used to test circuits. The final is available in a very big variety as far as a current rating and adjustable current rating is concerned.

FIGURE 2.20 shows an adjustable Power supply which uses a positive fixed voltage regulator. The regulator, which is used here, is a +5 V-regulator with an adjustable ground reference. This source produces an adjustable voltage higher than the 5 V of the regulator. P1 adjusts the ground reference of the regulator. If the voltage chop across $\boldsymbol{P_1V_G}$ = + 7 V, the output voltage will be $V_{out} = 7V + 5V = +12V$.



The same regulator can be used to produce an output Voltage lower than that of the regulator. The disadvantage of this circuit is that a positive as well as a negative power supply is required.

FIGURE 2.20 Regulator with voltage higher than fixed voltage of regulator

The potentiometer P_1 in **FIGURE 2.21** adjusts the ground reference of the regulator to a negative reference which causes the output voltage to drop to a level below the normal level.

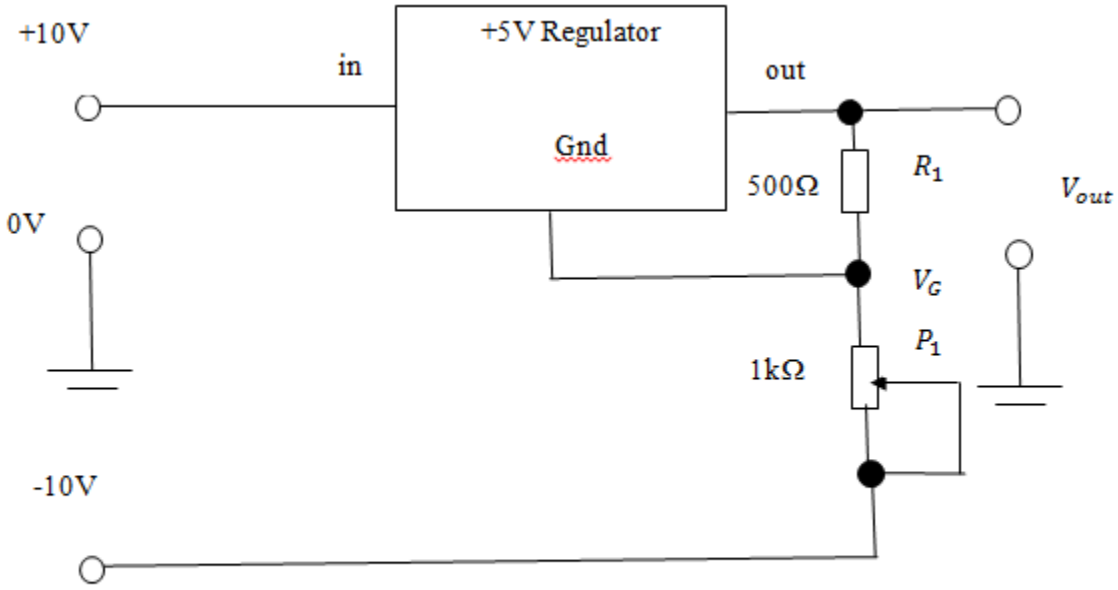


FIGURE 2.21 Regulator with voltage lower than fixed voltage regulator

If an output voltage of +1,5V is required V_G must be adjusted to -3,5V. A high stable, adjustable voltage regulator can be obtained if the feedback gain is increased with the aid of an amplifier.

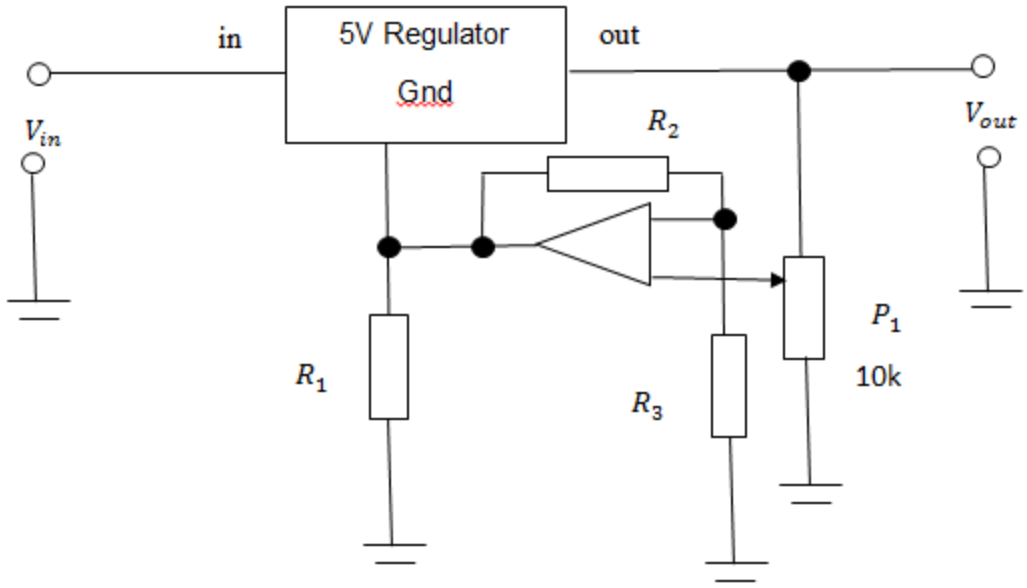


FIGURE 2.22 High stable adjustable voltage regulator

With a 5V regulator, this circuit will be able to produce an adjustable voltage of approximately: $+7\ to + 30V\ \leq (V_{in}\ - 2V)$.

2.13 INTEGRATED BRIDGE RECTIFIERS

An integrated bridge rectifier is an appliance of 4 diodes integrated into a single package. It is available in different sizes and is very useful.

FIGURE 2.24 depicts the circuit diagram and two types of integrated components available.

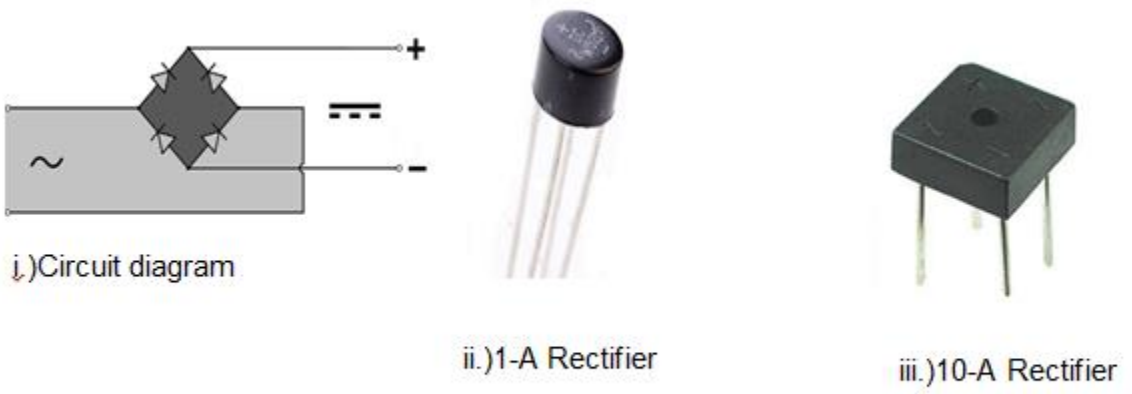


FIGURE 2.24

The advantages of the components are the amount of halved terminals and simplifying the outlay of a power supply a bit. If the component is damaged, it will cost more to replace than a single diode bridge circuit.

2.14 THREE-TERMINAL VOLTAGE REGULATORS

The three-terminal voltage regulators are integrated circuits with only three terminals; they are available in various voltage and current ratings and require no extra components. These regulators possess internal current limitations for overcurrent protection and terminal derivation, which make them indestructible to a certain extent. If sufficient heat sinking is supplied, specific regulators can produce more than 1A.

Though they are fixed voltage regulators, they can be used to produce adjustable output voltages and currents with the required external components. The circuits in **FIGURE 2.21 and FIGURE 2.22** are examples of these. Fixed voltage regulators are available in positive and negative output regulator units. The 7800 series are positive regulators and output voltages of 5; 6; 8; 8,5; 12; 15; 18 and 24V are obtainable. Oppositely the 7900- series are negative regulators with output voltages of -2;-5; -5,2;-6 ; -8; -12; -15; -18 and -24 V. ***In this syllabus the 7812-regulator is used***, FIGURE 2.25 depicts the integrated regulators.

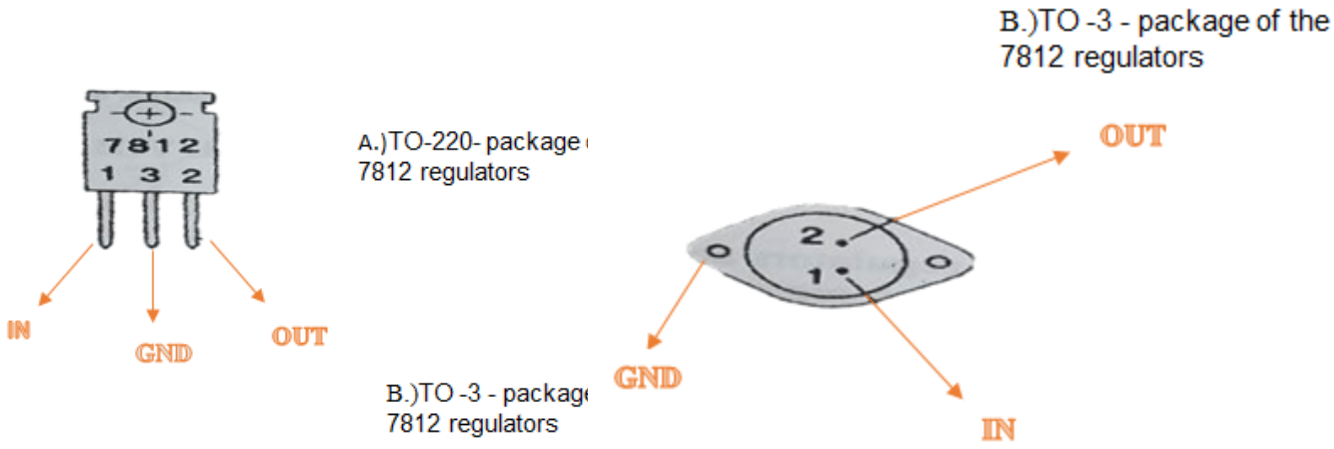


FIGURE 2.25

The 7812-regulator produces a positively regulated voltage of 12V and an output current of 1,5A for the TO – 220-package and TO-3-packages. A maximum voltage of 35V may be applied to the supply. The 78H12 – regulator, can maintain an output current of 5A. **FIGURE 2.26** shows how the voltage regulator is connected to a power supply.

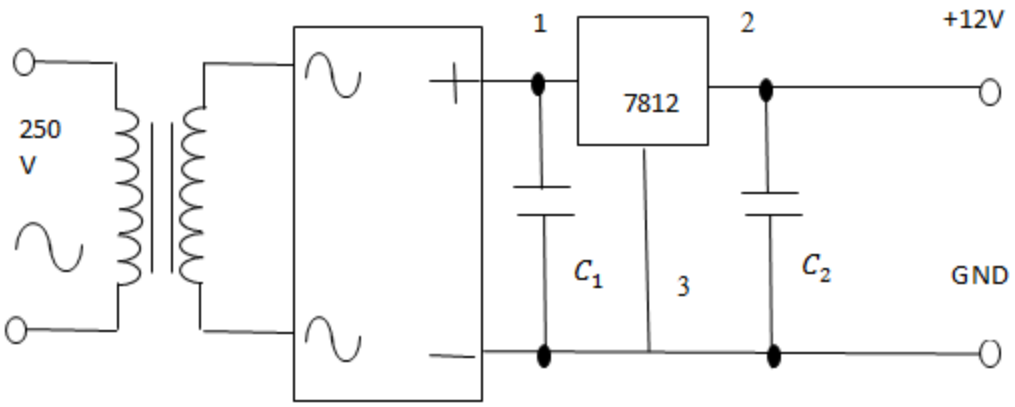


FIGURE 2.26 Connection of the 7812 - regulator

SELF CHECK 2

1.)The following information in connection with R-C π - filter circuit is known: a 50 Hz supply rectified by a bridge rectifier, RL= 1,5kΩ, C1=470µF, C2=47µF, R=10Ω$V_{dc} = 16V\ and\ V_{r(rms)} = 1{,}8V\ at$ C1.

1.1 Calculate the DC - Voltage across the load RL
1.2 Calculate the ripple voltage across the second capacitor C2
1.3 Calculate the ripple factor across the first capacitor C1
1.4 Calculate the ripple factor across the second capacitor C2.

2.)An LC π filter has a 0,5H inductor with a DC – resistance of 12Ω. The value of capacitor C2 is 100µF, and the value of RL is 2kΩ V_{dc}across C1 is 12V and $V_{r(rms)}$ Across C1 is 1,5V, f=50Hz.

2.1.) Calculate $V^1{}_{dc}$ across the load
2.2.) Calculate $V^1{}_{r(rms)} across\ C2$
2.3.) Calculate r^1 across C2

3.)Calculate the value of the capacitor C in FIGURE2.4 (a) if RL= 12Ω and $V_{dc=}15V$.
Assume that a full-wave rectifier is used, the line frequency f=50Hz and $V_{r(rms)} = 0{,}2V$.

4. Calculate the value the capacitor in FIGURE 2.4 (a) should have if the circuit faces a full-wave rectified sine wave to a ripple percentage of 3%. The line frequency is 50 Hz, and the load resistance is 500Ω

5. Calculate the value of RL in FIGURE 2.4 (a) if V_m =15 V, f =50 Hz after half-wave rectification, C=470µF and $I_{dc=}$ 10 mA.

6. Calculate the ripple percentage of the filter in FIGURE 2.4 (a) ifV_m = 16V, I_{dc} = 10 mA, f = 100 Hz after
full wave rectification and C=1000µF.

7. The following values of the circuit in FIGURE 2.4 (c) are known: $V_{dc=}$14 V,I_{dc} =1A,R =10Ω RL= 1 kΩ and $V^1{}_{r(rms)}$= 0,8 V.
Calculate :
7.1. The output DC-voltage
7.2 . The output ripple factor

8.)Calculate the value of the output DC voltage of the LC π -filter circuit in FIGURE 2.4 (d) if R1 = 5Ω,
RL=200 Ω and the DC - voltage across C1=18V.

9.)Calculate the value of the coil which should be turned for the filter circuit in FIGURE 2.4 (d) if the ripple voltages across C1 and C2 are 2,2 V and 0,8V, respectively, while the line frequency before full-Wave rectification is 50Hz and the capacitors each has a value of 100µF.

References

1. DJ Van Deventer, D & N Uitgewers (1991) Industrial Electronics N5 Edition, P.O Box 49649 Hercules .0030, Proud Enterprises First edition, First impression 1991 Publishers. Available from: https://www.bereatech-my.sharepoint.com
   [accessed 21st January 2021]

2. Johann Kraft, Future managers (2024), Industrial Electronics N5, 1st edition
3. C Thobejane,Troupant Publishers (2024) Industrial Electronics N5 1st edition, Northcliff
4. Available from: https://techterms.com [accessed 1st February 2021]
5. Available from: https://youtu.be/IjJWWGPjc-w [accessed 1st February 2021]